



MT-FlowFormer: A Semi-Supervised Flow Transformer for Encrypted Traffic Classification

Ruijie Zhao*
Shanghai Jiao Tong University
Shanghai, China
ruijiezhao@sjtu.edu.cn

Xianwen Deng*
Shanghai Jiao Tong University
Shanghai, China
2594306528@sjtu.edu.cn

Zhicong Yan
Shanghai Jiao Tong University
Shanghai, China
zhicongy@sjtu.edu.cn

Jun Ma
Shanghai Jiao Tong University
Shanghai, China
jun18217525728@163.com

Zhi Xue†
Shanghai Jiao Tong University
Shanghai, China
zxue@sjtu.edu.cn

Yijun Wang†
Shanghai Jiao Tong University
Shanghai, China
ericwyj@sjtu.edu.cn

ABSTRACT

With the increasing demand for the protection of personal network meta-data, encrypted networks have grown in popularity, so do the challenge of monitoring and analyzing encrypted network traffic. Currently, some deep learning-based methods have been proposed to leverage statistical features for encrypted traffic classification, which are barely affected by encryption techniques. However, these works still suffer from two main intrinsic limitations: (1) the feature extraction process lacks a mechanism to take into account correlations between flows in the flow sequence; and (2) a large volume of manually-labeled data is required for training an effective deep classifier. In this paper, we propose a novel semi-supervised framework to address these problems. To be specific, an efficient classifier with attention mechanism is proposed to extract features from flow sequences with low computational cost. Then, a Mean Teacher-style semi-supervised framework is adopted to exploit the unlabeled traffic data, where a spatiotemporal data augmentation method is designed as the key component to explore the spatial and temporal relationship within the unlabeled traffic data. Experimental results on two real-world traffic datasets demonstrate that our method outperforms state-of-the-art methods with a large margin.

CCS CONCEPTS

• **Networks** → **Network management**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

traffic classification, transformer, semi-supervised learning, data augmentation

*Equal contribution.

†Corresponding authors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

KDD '22, August 14–18, 2022, Washington, DC, USA

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9385-0/22/08...\$15.00

<https://doi.org/10.1145/3534678.3539314>

ACM Reference Format:

Ruijie Zhao, Xianwen Deng, Zhicong Yan, Jun Ma, Zhi Xue, and Yijun Wang. 2022. MT-FlowFormer: A Semi-Supervised Flow Transformer for Encrypted Traffic Classification. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)*, August 14–18, 2022, Washington, DC, USA. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3534678.3539314>

1 INTRODUCTION

With the rapid development of the Internet, traffic classification has become a vital task in cyberspace security and network management [2, 13, 19, 20, 24]. Many content and service providers use traffic classification technology to improve network performance and provide their users with high quality service and experience [26, 28]. However, with the widespread adoption of network encryption technology, traditional rule-based solutions become more and more ineffective due to the inaccessibility to the raw content of encrypted data packets. Thus, it is necessary to adopt new mechanisms to effectively classify encrypted traffic.

In recent years, the identification of traffic generated by the encrypted tunnels of virtual private network (VPN) and anonymity network has attracted much attention, and machine learning (ML) or deep learning (DL) algorithms that deal with the robust statistical features of traffic flows have been widely acknowledged as the mainstream method. In addition, we note that many studies have shown that most traffic data has obvious temporal correlations [34, 38]. For example, when a user uses an anonymity network to browse the web, many related application requests will appear in a short period. These application requests will generate corresponding multiple flows in the traffic. Based on the temporal correlation, some researchers [28, 39, 42] proposed to use flow sequences for traffic analysis. While leveraging flow sequences can help improve classification accuracy, the feature extraction process lacks a mechanism to effectively discriminate the correlation and importance of different flows in the flow sequence. Moreover, the real-world traffic is often mixed with unimportant or irrelevant flows in the flow sequence. As a result, the features extracted from the mixed flows deteriorate classification performance. In addition, most studies are conducted within the supervised learning framework, which means that a large volume of manually-labeled real traffic data is required to learn an effective classifier. Compared with labeled encrypted traffic data, we could easily collect abundant unlabeled encrypted

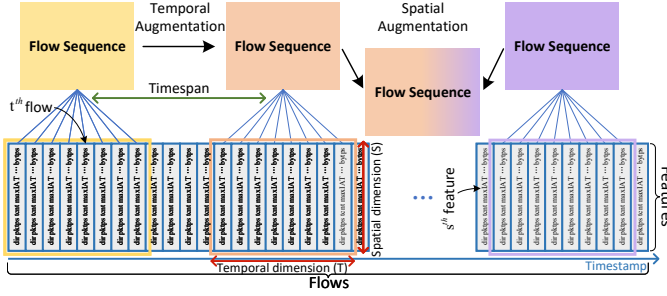


Figure 1: Data augmentations for our task according to spatiotemporal correlation of the traffic. The model input is the flow sequence of fixed size $T \times S$ for temporal $\times$ spatial dimensions. Timespan represents the temporal distance between two flow sequences.

traffic data in real applications. Obviously, it is highly desirable to leverage additional unlabeled data for improving the classification performance.

To address the challenges mentioned above, we develop a semi-supervised framework (MT-FlowFormer) for encrypted traffic classification. We first build a Transformer-based baseline model for encrypted traffic classification, denoted as FlowFormer, which leverages an unique self-attention mechanism to allow interaction between related flows in the flow sequence. Since traffic classification models are usually deployed on edge devices (e.g., router and switch) with limited storage and computing capabilities [22, 41], our model adopts an efficient structure for feature extraction. Concretely, to reduce the model complexity, FlowFormer gradually reduces the flow sequence length of the hidden representation during the feature extraction process. Besides, we apply a hierarchical structure where the key idea is to capture both shallow-layer features and deep-layer features. By aggregating the information from different layers, our classifier can combine both local and global attention.

To incorporate unlabeled traffic data into training, we adopt a Mean Teacher-style semi-supervised learning framework to train the model. Mean Teacher has been successfully adapted to various semi-supervised tasks, such as image segmentation [7], object detection [5], and behavior analysis [35]. However, the success of Mean Teacher on various image-related tasks largely resorts to the combination of a diversity of image augmentation methods and consistency regularization exerted on the augmented unlabeled images. To adapt this mature semi-supervised paradigm into traffic data classification, we adopt two data augmentation methods specialized for encrypted traffic data as shown in Fig. 1: (1) according to the temporal correlation, take the flow sequence with a timespan (i.e., temporal distance) from the original flow sequence, and (2) according to the spatial correlation, use the fusion algorithm to change the data content. The prediction of the augmented flow sequence from the student network is encouraged to be consistent with the prediction of the raw flow sequence from the teacher network.

Our main contributions can be summarized as follows:

- We develop an efficient classifier (FlowFormer) to classify encrypted traffic by leveraging attention mechanism on the flow sequence. FlowFormer significantly reduces the model complexity by gradually compressing the sequence length

of hidden states, and captures both shallow-layer features and deep-layer features by aggregating the information from different layers.

- We apply a Mean Teacher-style semi-supervised learning framework (MT-FlowFormer) based on consistency regularization to leverage a few labeled data and many unlabeled data for training a high performance classifier. Moreover, a spatiotemporal data augmentation method is adopted as the key component to explore the temporal and spatial relationship within the unlabeled traffic data.
- Our MT-FlowFormer outperforms the state-of-the-arts by a large margin on two real-world encrypted traffic datasets, which verifies that the proposed baseline model can efficiently extract features from flow sequences and the proposed framework can effectively leverage additional unlabeled data to boost the traffic classification performance.

2 RELATED WORK

DL-Based Methods. The identification of encrypted traffic is a typical traffic classification task. Network traffic classification methods can be divided into three types (i.e., rule-based, ML-based and DL-based) according to their implementation. The rule-based analysis method can only analyze the uncomplicated network traffic, and the port number identification has been proved to be invalid [21] in the complex network environment.

To solve the issues with rule-based analysis methods, researchers applied to use payload data and statistical features, both of which are more suitable for identifying specific application scenarios in complex traffic. However, affected by the traffic encryption technology, only the first few packets that contain handshake information are typically unencrypted, resulting in limited information for analysis in the payload [26, 28]. In the early stage, the statistical features of encrypted traffic were analyzed based on ML algorithms [17, 23].

Recently, benefiting from the powerful feature extraction capabilities of DL algorithms, many researchers have proposed DL-based traffic identification methods with better performance. For instance, Yang *et al.* [40] proposed a scheme to extract encrypted traffic features by convolutional neural networks (CNN), and its classification performance is significantly improved compared to traditional machine learning algorithms. Liu *et al.* [19] proposed a multi-layer encoder-decoder structure which can mine the potential sequential characteristics of flows deeply. Zheng *et al.* [47] adopted a meta-learning approach to learn representative features from the raw flows and then classifies them in a unified framework.

Although these DL models have achieved great success in the task of traffic classification, their feature extraction process ignores the importance of different flows in the flow sequence, which deteriorate the classification performance. Transformer, as the most effective deep learning model with attention mechanism, is very suitable for learning and leveraging the importance of different flows in the flow sequence for feature extraction, but the high complexity of the traditional Transformer model hinders the deployment on edge devices. Thus, we build an efficient classifier (FlowFormer) for encrypted traffic classification, which has excellent performance and greatly reduces the model complexity. The specific structure of our classifier can be seen in Section 4.1.

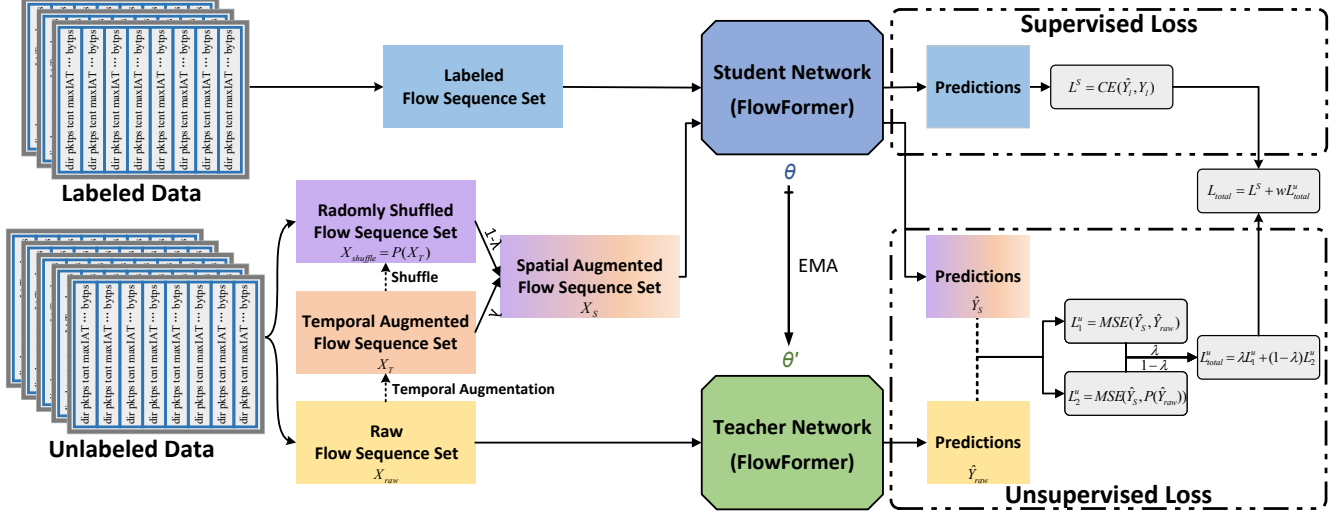


Figure 2: The schematic illustration of our MT-FlowFormer framework. We first build an efficient classifier (FlowFormer; see Fig. 3) to classify encrypted traffic by leveraging attention mechanism on the flow sequence. After that, we train our FlowFormer with a few labeled data and many unlabeled data by our framework; see Section 4.2 for details. Note that the sample space of the shuffled flow sequence set is the same as that of temporal augmented flow sequence set.

Semi-Supervised Methods. In the task of traffic classification through deep learning, obtaining a large amount of labeled data is a cumbersome and time-consuming process. Therefore, many researchers have begun to take advantage of unlabeled network traffic data through semi-supervised learning (SSL), which is more suitable for practical applications. Rezaei *et al.* [27] pre-trained models on unlabeled dataset by fitting statistical features and then transferred the learned weights to a new model that is re-trained on labeled dataset. Iliyasu *et al.* [18] proposed a semi-supervised learning approach using deep convolutional generative adversarial network (DCGAN). Aouedi *et al.* [3] applied stacked sparse autoencoder (SSAE) to construct a semi-supervised model, and pre-trained the network by reconstructing unlabeled data. Fahad *et al.* [11] implemented semi-supervised approach through clustering and label propagation, and experiments were carried out using sixteen network traffic datasets. At present, SSL in traffic classification scenarios mainly rely on clustering, label propagation, and generative models, which often have problems such as insufficient feature extraction and unstable training.

Consistency regularization is the current research hotspot of SSL. There are many excellent frameworks for this method, which have been applied in a wide range of scenarios [5, 7, 14]. Mean Teacher [11] averages model weights to form a target-generating teacher model, which has huge advantages in large datasets and online learning. FixMatch [29] is another advanced method for SSL, which has achieved state-of-the-art performance across a variety of standard SSL benchmarks by combining consistency regularization and pseudo-labeling.

To improve the performance of SSL in traffic classification scenario, a consistency regularization scheme based on the spatiotemporal features of flow sequences is proposed in our method, which is introduced in Section 4.2.

3 PRELIMINARIES

In this section, we first give the definition of the encrypted traffic classification problem. Then, the flow generation process and the Transformer network are introduced briefly.

3.1 Problem Definition

We consider the problem of encrypted traffic classification as semi-supervised classifier learning. Our task is to leverage a few labeled samples and many unlabeled samples for training a classifier with excellent performance. To verify the performance and generality of our classifier for encrypted traffic, we selected two typical encrypted traffic networks (i.e., virtual private networks and anonymity networks) for analysis. There are three datasets in each encrypted traffic network: a labeled training set, an unlabeled training set, and a test set. In our experiment, the test set has fixed sample space, but the sample space of the labeled training set and unlabeled training set are determined according to the experimental setting.

Note that the raw traffic flow is time series data, which means that the data has spatiotemporal relationship between flows [42]. Combining the spatiotemporal relationship of the traffic, we design our baseline model and semi-supervised framework. Specifically, our baseline model is not to analyze a flow but a flow sequence composed of multiple continuous flows. Although many studies [28, 39, 42] have performed traffic analysis on flow sequences and proved to significantly improve classification performance, they didn't focus on the temporal correlations between flows. Therefore, we use the self-attention mechanism of Transformer to allow the interaction between related flows. Moreover, we adopt a hierarchical structure to extract multiscale features. The data augmentation strategy of our semi-supervised framework also takes advantage of the spatiotemporal relationship, which is applied to unlabeled data. We select temporal augmentation data

within the timespan, and the selected data cannot exceed the data boundary determined by the five-tuple (i.e., source/destination IP addresses, source/destination port numbers, and protocol). In addition, the data content of each flow is the statistical features of the raw traffic, so we use the MixUp-based spatial augmentation to mix the statistical features of different flows. Specific data augmentation methods are introduced in Section 4.2.

3.2 Flow Generation

In packet switching networks, traffic flow is a sequence of packets carrying information between two hosts. The raw traffic data is captured and stored in pcap (abbreviation of packet capture) format. The pcap file mainly contains three parts (i.e., pcap header, packet header, and packet data). Although the pcap file has complete information, it cannot reflect some statistical features of the flow. Besides, encryption technology makes it difficult to directly read the complete information of the pcap file of encrypted traffic. Thus, it is necessary to calculate and extract the statistical features to enrich the feature information of each flow.

For traffic mining, the preprocessing and proper mathematical transformation to find invariances are the basis for building a classifier which performs well in practice. Our method adopts Tranalyzer2¹, a lightweight flow generator and packet analyzer, to extract the statistical features of the raw traffic. For flows with a long duration, we segment them according to the maximum duration setting to better meet the real-time requirements of traffic classification. The extracted features of each flow include a total of 84 statistical features (e.g., packet length, inter-arrival time, and flow duration). Since the extraction of statistical features is not the contribution of this paper, more detailed information about statistical features can be found in the instructions on the official website of Tranalyzer2. To preserve the spatiotemporal correlations of the raw traffic, we store the generated flow sequentially according to the timestamp during extraction. Finally, we take continuous flows as a flow sequence for encrypted traffic classification, where each flow contains a total of 84 features.

3.3 Transformer

Transformer [31] was first proposed in machine translation tasks, which is mainly composed of multi-head self-attention modules and position-wise feed-forward networks (FFN). Subsequently, the Transformer network has been widely used in various tasks of natural language processing (NLP) and computer vision (CV) due to its excellent performance, and many variant networks [9, 25] have been produced. Researchers' exploration of Transformer mainly includes two aspects:

- Module-level improvements such as sparse attention [6], improved multi-head [15], and implicit representations [33].
- Architecture-level improvements such as strengthening cross-block connectivity [12], adaptive computation time [10], and lightweight model [8, 37].

The Transformer network structure is composed of an encoder and a decoder to solve typical Seq2Seq problems. Transformer can also use the encoder or the decoder separately, which is suitable

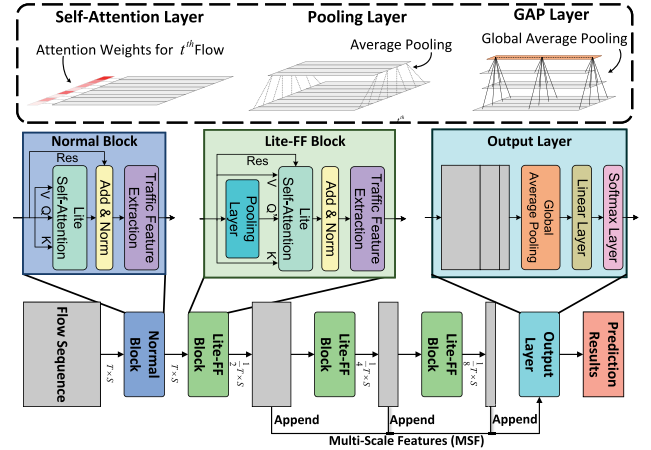


Figure 3: The schematic illustration of FlowFormer, which is composed of one Normal Block and three Lite-FF Blocks. Multi-scale features (MSF) are extracted by these Lite-FF Blocks. Then Global Average Pooling (GAP) layer is used to aggregate these features. Finally, a simple fully-connected layer is leveraged to predict the classification of encrypted traffic.

for classification tasks or sequence generation tasks, respectively. In Transformer, its self-attention mechanism can connect different positions in the sequence through the Query-Key-Value model to calculate the feature representation of the entire sequence. Relying on this design, Transformer has the ability to capture the internal correlation of features. Inspired by the excellent Transformer-based studies mentioned above, we design an efficient classifier based on Transformer for traffic classification.

4 METHODOLOGY

The proposed classifier (FlowFormer) and the semi-supervised learning framework (MT-FlowFormer) are described in this section.

4.1 FlowFormer

We present FlowFormer, an efficient classifier based on Transformer, as shown in Fig. 3, to effectively classify encrypted traffic.

Taking the Normal-Transformer Block (Normal Block) into account, suppose the input is a flow sequence with a fixed size of $T \times S$, corresponding to the temporal and spatial dimensions, respectively. The most important structure of Normal-Transformer Block is multi-head self-attention modules. As mentioned above, the input $X \in \mathbb{R}^{T \times S}$. Every self-attention head computes Q, K, V by linear transformation, representing Query Matrix, Key Matrix, and Value Matrix, respectively.

$$Q = XW^Q, K = XW^K, V = XW^V, \quad (1)$$

where $W^Q, W^K, W^V \in \mathbb{R}^{S \times d}$ are trainable parameters, d is the hidden layer dimension, and $Q, K, V \in \mathbb{R}^{T \times d}$ represent Query Matrix, Key Matrix, and Value Matrix, respectively. Then the output of this self-attention head is computed:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V. \quad (2)$$

¹<https://tranalyzer.com/>

Matrix calculation can be parallel and save computation time and space, but it also brings obstacles to understand the essence of self-attention. More specifically, Query Matrix, Key Matrix, and Value Matrix all have T vectors, equal to time-series length. For every query vector q , calculate its similarity with each key vector k as scores, to perform a weighted summation on all value vectors v . Therefore, the length of the output sequence depends on the number of query vectors. In Normal-Transformer Block, the length of output is always T , making the calculation complexity of Transformer very high. We only use one Normal-Transformer Block as the first layer.

To reduce the model complexity, we adopt Lite FlowFormer Block (Lite-FF Block) to reduce the sequence length of the output by half. Formally, taking the same $X \in \mathbb{R}^{T \times S}$ as input, we first use a Average Pooling layer to reduce the sequence length by half:

$$X' = \text{Pooling}(X), \quad (3)$$

where $X' \in \mathbb{R}^{\frac{T}{2} \times S}$. Then Query Matrix is computed from X' , while Key Matrix and Value Matrix remain the same:

$$Q' = X'W^Q, K = XW^K, V = XW^V, \quad (4)$$

where $Q' \in \mathbb{R}^{\frac{T}{2} \times d}$, $K, V \in \mathbb{R}^{T \times d}$. The computation of self-attention is similar:

$$\text{Attn}(Q', K, V) = \text{softmax}\left(\frac{Q'K^T}{\sqrt{d}}\right)V. \quad (5)$$

Because the sequence length of output is equal to the number of query vectors, Lite-FF Block eventually reduce it by half, which can effectively reduce computational complexity. In addition, the number of self-attention heads in both Lite-FF Block and Normal Block is set to 1, to further reduce model complexity.

By stacking three Lite-FF Blocks, the sequence length of output gradually decreases and multi-scale features are extracted by the hierarchical structure. In order to avoid complicated calculations, these multi-scale features are directly compressed into a one-dimensional vector through global average pooling (GAP). Then a simple fully-connected layer is used to output the classification results of encrypted traffic.

4.2 Semi-Supervised Learning Based on Mean Teacher

Mean Teacher has achieved great success in image classification by utilizing a large number of unlabeled data. In this semi-supervised framework, there are two networks with same structures, namely teacher and student model. The core of Mean Teacher is to encourage the consistent predictions of teacher and student model for different views of the same image. Thus, data augmentation methods are critical to the performance of Mean Teacher.

There are many data augmentation methods for images, such as color jittering, random resizing, and rotation. However, these augmentation methods cannot be applied to statistical features extracted from encrypted traffic, which are essentially tabular data. Inspired by some effective semi-supervised training frameworks [4, 30, 32], we deploy our semi-supervised training paradigm. Besides, we present two traffic data augmentation methods, according to temporal correlation and spatial correlation, respectively.

4.2.1 Temporal Augmentation. Let $\mathcal{D} = \{x_i : i \in (1, \dots, N)\}$ be the training set. Note that they are not shuffled, so the smaller the distance between the samples, the closer the timestamps between the flow sequences. Formally, given a x_i , we randomly choose a flow sequence $x_i^{(t)}$ whose temporal distance from x_i is less than timespan τ :

$$x_i^{(t)} = \text{Random}(x_j : i - \tau \leq j \leq i + \tau). \quad (6)$$

4.2.2 Spatial Augmentation. Spatial augmentation is based on linear interpolation of different samples. The definition of linear interpolation is as follows:

$$\text{Mix}_\lambda(a, b) = \lambda \cdot a + (1 - \lambda) \cdot b. \quad (7)$$

Taking two samples randomly: (x_i, y_i) and (x_j, y_j) , it's believed that the linear interpolation of them should maintain the properties of both at the same time, which means linear interpolation sample $\text{Mix}_\lambda(x_i, x_j)$ should correspond to linear interpolation of labels $\text{Mix}_\lambda(y_i, y_j)$.

MixUp [44] is widely used in supervised learning and semi-supervised learning to alleviate overfitting by taking $\text{Mix}_\lambda(y_i, y_j)$ as labels of $\text{Mix}_\lambda(x_i, x_j)$. In semi-supervised scenarios, since there is no available label for unlabeled data, MixMatch [4] and ICT [32] propose to use the model prediction as labels. Inspired by these two methods, we develop the spatial augmentation method by mixing the output of the teacher model as labels.

4.2.3 MT-FlowFormer. In this section, we give a complete illustration of MT-FlowFormer. Both student and teacher model directly employ the structure of FlowFormer $f : x \rightarrow y$. Denote the weights of student and teacher model as θ and θ' , respectively. Thus $f(x, \theta)$ and $f(x, \theta')$ means to make predictions with student and teacher model, respectively.

Randomly sample a batch containing n samples from $\mathcal{D}$, denoted as X_{raw} . First, make predictions on it with teacher model, and take the results as labels:

$$\hat{Y}_{raw} = f(X_{raw}, \theta'), \quad (8)$$

by this way, we solve the problem that MixUp is difficult to use for unlabeled data, by providing labels with teacher model.

Then, we perform temporal augmentation on each sample in X_{raw} , and obtain the temporal augmented batch X_T , by randomly selecting an adjacent flow sequence for each sample within timespan τ .

For effectiveness, we use MixUp between samples of the same batch X_T to achieve spatial augmentation. Formally, a random permutation P is leveraged to shuffle X_T :

$$X_{shuffle} = P(X_T). \quad (9)$$

Then, spatial augmentation is performed by mixing X_T with $X_{shuffle}$:

$$X_S = \text{Mix}_\lambda(X_T, X_{shuffle}), \quad (10)$$

where X_S is actually the spatiotemporal augmentation from X_{raw} , and we make predictions on it with student model:

$$\hat{Y}_S = f(X_S, \theta). \quad (11)$$

Table 1: The performance comparison with other models using different labeled sample sizes.

Model	SJTU-AN21						ISCXVPN2016						FLOPs	Params
	0.1%	0.5%	1%	5%	10%	100%	0.1%	0.5%	1%	5%	10%	100%	(10^6)	(10^4)
1D-CNN	17.7%	32.6%	49.9%	50.1%	51.1%	53.8%	38.3%	56.9%	70.5%	82.2%	85.8%	91.8%	22.64	661.7
2D-CNN	22.0%	32.7%	50.8%	52.0%	52.1%	53.7%	47.2%	55.7%	61.0%	78.4%	80.6%	88.5%	13.88	327.5
3D-CNN	23.7%	33.4%	46.6%	50.4%	50.7%	52.8%	41.7%	43.4%	55.0%	71.9%	81.0%	91.5%	1.42	44.7
ACNN	23.5%	65.8%	68.3%	78.6%	80.7%	81.0%	66.1%	82.6%	87.6%	88.2%	88.7%	91.2%	6.72	42.3
LSTM	52.4%	76.1%	78.7%	79.0%	80.9%	81.3%	76.0%	82.8%	85.9%	89.0%	89.4%	92.5%	4.10	50.7
AttnLSTM	41.9%	75.0%	79.0%	79.6%	80.0%	81.2%	68.3%	79.4%	87.5%	88.8%	89.5%	92.2%	3.97	26.5
FS-Net	45.0%	68.7%	72.5%	80.0%	80.2%	83.7%	67.3%	78.9%	87.2%	87.5%	87.7%	92.4%	6.22	64.6
NormalFormer	42.8%	65.7%	73.8%	80.3%	81.9%	86.1%	65.7%	75.8%	86.2%	88.9%	90.7%	92.5%	4.46	69.6
Ours (FlowFormer)	60.3%	80.6%	81.0%	84.0%	84.7%	86.8%	77.1%	87.4%	88.3%	89.6%	90.5%	93.2%	1.07	26.2

Unsupervised loss is computed by comparing $\hat{Y}_S$ with $\hat{Y}_{raw}$, according to mean square error (MSE):

$$\mathcal{L}_u = \lambda \cdot \text{MSE}(\hat{Y}_S, \hat{Y}_{raw}) + (1 - \lambda) \cdot \text{MSE}(\hat{Y}_S, P(\hat{Y}_{raw})), \quad (12)$$

where P is the same permutation used for shuffle X_T , making the labels $P(\hat{Y}_{raw})$ correspond to shuffled samples.

Then supervised loss is computed with labeled data through cross-entropy loss:

$$\mathcal{L}_s = \text{CE}(f(X_l, \theta), Y_l). \quad (13)$$

The total loss of MT-FlowFormer is:

$$\mathcal{L}_{total} = \mathcal{L}_s + w \mathcal{L}_u, \quad (14)$$

where w is a scalar hyperparameter. The total loss is used to update the weights of only student model through gradient descent algorithm, then the weights of teacher model are updated through exponential moving average (EMA) strategy:

$$\theta' = \alpha \theta' + (1 - \alpha) \theta, \quad (15)$$

where $1 - \alpha$ is the step size, and usually set to 0.001.

5 EXPERIMENT AND RESULTS

5.1 Experiment Settings

5.1.1 Data Preparation. Our experiments are conducted on two real-world encrypted traffic datasets SJTU-AN21 dataset [46] and ISCXVPN2016 dataset [16], respectively.

SJTU-AN21 dataset is collected in the latest version of the three most popular anonymity networks (Tor, I2P, JonDonym). In this dataset, Eepsites, IRC, Snark, and Video are the four main anonymity services in the I2P network, and Browsing, Chat, FTP, Streaming, and BitTorrent are the five main anonymity services in the Tor network. This dataset has independent training dataset and test dataset, which have 29,214 flows and 6,979 flows respectively.

ISCXVPN2016 dataset is used to evaluate the effectiveness of encrypted traffic classification on virtual private networks. The traffic was generated through an external VPN service provider with OpenVPN (UDP mode). A total of 7 application services are included in the selected dataset and 28,395 flows are parsed for classification. Finally, this selected dataset is divided into independent training

dataset and test dataset according to the proportion of 80% and 20% for each class.

5.1.2 Implementation Details. In the training phase, we train the model with a stochastic gradient descent (SGD) optimizer. The learning rate is set to 0.025 and a weight decay of 0.0003 is used to optimize the whole network with 300 epochs. The weight of the unsupervised loss w is 10. The impacts of different timespans and distribution parameters are discussed in Section 5.4. The proposed approach is implemented using PyTorch 1.9.0 and trained on PC with Intel® Core™ i9-11900K@3.50 GHz, 64 GB RAM, and an NVIDIA GeForce RTX3090 GPU.

5.2 Evaluation on Encrypted Traffic

5.2.1 Comparison with Other Models. To comprehensively evaluate our classifier, we compare our model with a range of baselines and state-of-the-art models. We use the same semi-supervised framework to train each model. Floating-Point-Operations (FLOPs) and Parameters are introduced to reflect model complexity, and the accuracy is used to evaluate classification performance. All comparison models are listed:

- **1D-CNN [43], 2D-CNN [36], and 3D-CNN [45]** are the payload-based CNN models for feature extraction, which directly analyze the payload data and header information.
- **ACNN [40]** analyzes the statistical features of traffic using CNN and 3-layer MLP for encrypted traffic classification.
- **LSTM [1] AttnLSTM [42], and FS-Net [19]** are the traffic classification methods based on the LSTM network. AttnLSTM [42] introduced attention mechanism to score the importance of each flow. FS-Net [19] leveraged reconstruction loss to ensure that the extracted features by LSTM contain all the information of the flow sequences.
- **NormalFormer** is composed of 4 Normal Blocks, which can be considered as a normal version of FlowFormer.

As is shown in Table 1, our method with only 5% of labeled data on SJTU-AN21 dataset can achieve better classification performance than the state-of-the-art methods using 100% of labeled data. The payload-based models show poor classification capabilities, because the information for analysis is limited by the traffic encryption technology. ACNN, LSTM, and FS-Net cannot find the

Table 2: The performance comparison with other semi-supervised learning methods.

Method	SJTU-AN21					ISCXVPN2016				
	0.1%	0.5%	1%	5%	10%	0.1%	0.5%	1%	5%	10%
Supervised	45.1%	66.3%	69.0%	79.6%	81.1%	66.0%	80.0%	84.5%	89.0%	90.3%
Label Propagation [48]	31.3%	44.1%	45.0%	50.9%	52.3%	62.6%	69.4%	72.1%	77.0%	80.4%
DCGAN [18]	45.1%	64.3%	71.5%	78.2%	78.5%	69.9%	78.7%	82.4%	89.1%	89.5%
Original Mean Teacher [30]	48.2%	67.7%	70.4%	80.5%	81.3%	67.1%	80.3%	84.7%	89.2%	90.3%
FixMatch [29]	58.5%	70.0%	74.4%	80.7%	82.7%	68.3%	82.2%	86.4%	89.3%	90.4%
Ours	60.3%	80.6%	81.0%	84.0%	84.7%	77.1%	87.4%	88.3%	89.6%	90.5%

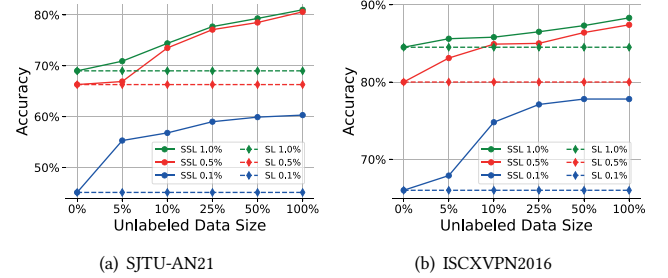
correlations between flows, which limits their ability to classify traffic. NormalFormer uses an effective self-attention mechanism to allow interactions between related flows, even if they are far away. Therefore, it can obtain a higher accuracy rate with enough labeled data. However, NormalFormer cannot fully learn temporal correlations between flows when the number of labeled data is small, which makes its accuracy rate even worse than LSTM-based models. What’s more, its high complexity makes it unsuitable for deployment in network devices. In addition, we can observe that as the size of labeled data increases, our method suffers a “ceiling effect” on both SJTU-AN21 and ISCXVPN2016 datasets (i.e., as the performance gets closer to the ceiling, the improvement becomes smaller).

5.2.2 Comparison with Other Semi-Supervised Frameworks. To demonstrate the proposed framework’s prominent utilization of unlabeled traffic data, we reimplement several other semi-supervised traffic classification methods, including both traditional and deep learning-based SSL methods. Specifically, we employed Label Propagation [48] as the baselines for traditional method. We also experiment two deep learning-based semi-supervised frameworks (DCGAN [18] and FixMatch [29]), which have been investigated in previous semi-supervised traffic classification works. For fair comparison of different semi-supervised frameworks, we reimplement Original Mean Teacher and FixMatch with the proposed FlowFormer model. Noted that the Original Mean Teacher is different from our method in that it does not apply the data augmentation. Then we carefully adjust hyperparameters and report the best results on these frameworks. The results are shown in Table 2.

As shown in Table 2, the results of the proposed spatiotemporal consistent learning framework surpass all other semi-supervised frameworks despite the same network architecture being employed. The label propagation achieves poor results, due to the traffic statistics being extremely noisy and chaotic for traditional methods to handle. The DCGAN employs adversarial techniques to tighten the decision boundary for each class. However, the results are not satisfactory under the traffic classification scenario possibly because the generative adversarial networks cannot accurately capture the noisy distribution of traffic data. Because no data augmentation is applied in the Original Mean Teacher, the performance improvement is also very weak. FixMatch treats highly confident predictions of teacher model as pseudo labels to guide the training of student model. However, MixUp provides a smoother estimate of

Table 3: Ablating augmentations.

Model	SJTU-AN21			ISCXVPN2016		
	0.1%	0.5%	1%	0.1%	0.5%	1%
Ours	60.3%	80.6%	81.0%	77.1%	87.4%	88.3%
Ours w/o T	54.1%	78.6%	78.9%	73.5%	86.4%	86.8%
Ours w/o S	56.2%	76.4%	78.6%	74.3%	84.8%	86.8%
Ours w/o TS	48.2%	67.7%	70.4%	67.1%	80.3%	84.7%

**Figure 4: Accuracy for SJTU-AN21 and ISCXVPN2016 datasets with different unlabeled data size.**

uncertainty, limiting the training of FixMatch. Therefore, Mean Teacher using mean square error (MSE) could be more suitable than FixMatch for the proposed spatiotemporal consistent learning framework.

5.3 Ablation Study

5.3.1 Traffic Data Augmentations. To further examine the level of benefit that data augmentations bring to the performance, an ablation study is performed on MT-FlowFormer with a few labeled data and all unlabeled data. The results are shown in Table 3. We ablate temporal (T), spatial (S) and both temporal and spatial (TS) augmentations from our method. We make two observations:

- Either temporal augmentation or spatial augmentation can improve classification performance compared to no augmentation, which means that both augmentations are very effective for semi-supervised learning.

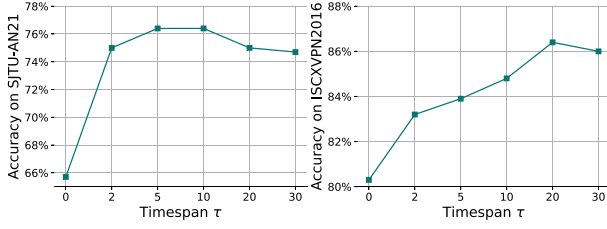


Figure 5: Accuracy for SJTU-AN21 and ISCXVPN2016 datasets with different timespan τ . Experiments are conducted with 0.5% labeled samples.

- We can observe that temporal augmentation and spatial augmentation can jointly promote training to achieve the best classification performance.

5.3.2 Size of Unlabeled Data. As an important factor affecting performance in the field of semi-supervised learning, the amount of unlabeled data should not be ignored. Thus, we evaluate our method with 0.1%, 0.5% and 1% labeled data of two datasets while varying the size of unlabeled data from 0, 5%, 10%, 25%, 50%, and 100%. The results are given in Fig. 4. We can observe that using the proposed MT-FlowFormer to exploit unlabeled data improves the performance especially when more unlabeled data are provided. In addition, after the unlabeled data size reaches 50%, the improvement brought by increasing the size of unlabeled data becomes smaller.

5.4 Discussions

In this section, we further investigate the impact of two hyperparameters in these two data augmentation methods, that are timespan τ and beta distribution parameter α in temporal and spatial dimension, respectively. To better reflect the effect of different hyperparameters for each augmentation, we only use one augmentation at a time. Besides, we discuss the impact of the weight of unsupervised loss.

5.4.1 Parameter τ for Temporal Augmentation. Timespan τ represents the temporal distance between two flow sequences. Fig. 5 shows the results of different maximum timespans for temporal augmentation. The longer the temporal distance means the stronger augmentation and the weaker correlation between the flow sequences. It can be observed that increasing the maximum timespan up to 10 or 20 is beneficial and beyond that performance decreases. Thus it can be concluded that both too short and too long temporal distance will weaken the effect of temporal augmentation.

5.4.2 Parameter α for Spatial Augmentation. When performing spatial augmentation, the interpolation ratio obeys the beta distribution $\beta(\alpha, \alpha)$. We sample interpolation ratio in it and only retain values greater than 0.5. In this case, the larger the α , the stronger the spatial augmentation. Fig. 6 shows the results of different values of α . Results show that stronger spatial augmentation will benefit classifying traffic, but if α is too large, the accuracy will drop a bit. This is because when α is greater than 200, all values sampled in beta distribution tend to be close to 0.5, which will limit the space of synthetic samples.

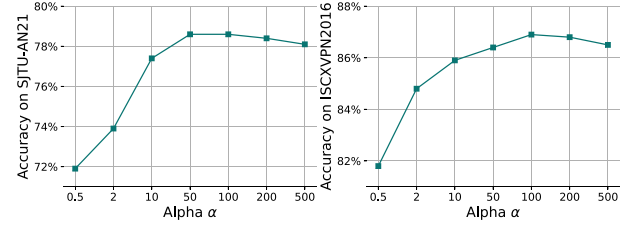


Figure 6: Accuracy for SJTU-AN21 and ISCXVPN2016 datasets with different α . Experiments are conducted with 0.5% labeled samples.

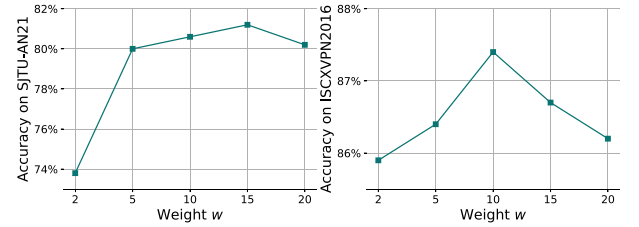


Figure 7: Accuracy for SJTU-AN21 and ISCXVPN2016 datasets with different unsupervised loss weight w . Experiments are conducted with 0.5% labeled samples.

5.4.3 Investigation on Unsupervised Loss Weight. We also investigate the impact of the weight of unsupervised loss w in our framework. To clearly show the impact of the w on performance, we focus on SJTU-AN21 and ISCXVPN2016 datasets with 0.5% labeled and 100% unlabeled data. Results are shown in Fig. 7. In fact, w controls the balance of supervised loss and unsupervised loss. A small w means paying more attention to labeled samples but ignoring unlabeled samples. Experiments prove that a weight of 10 is suitable for both two datasets.

6 CONCLUSION

In this paper, we propose a novel semi-supervised method based on Transformer and the consistency regularization for encrypted traffic classification. Our key idea is to build an efficient Transformer-based classifier (FlowFormer) with attention mechanism to efficiently extract deep features from flow sequences. We apply Mean Teacher-style semi-supervised framework to leverage additional unlabeled data for further improving classification performance, where temporal and spatial augmentations are adopted as the key component to explore the temporal and spatial relationship within the training traffic data. Experimental results on two real-world traffic datasets demonstrate that the method outperforms state-of-the-art methods with a large margin.

ACKNOWLEDGMENT

This work was supported by the Foundation Item: Cyber Security from the National Key Research and Development Program of Shanghai Jiao Tong University under Grant 2019QY0703.

REFERENCES

- [1] Sara A. Althubiti, Eric Marcell Jones, and Kaushik Roy. 2018. LSTM for anomaly-based network intrusion detection. In *International Telecommunication Networks and Applications Conference*. 1–3.
- [2] Blake Anderson and David A. McGrew. 2017. Machine learning for encrypted malware traffic classification: accounting for noisy labels and non-stationarity. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. 1723–1732.
- [3] Ons Aouedi, Kandaraj Piamrat, and Dhruvjyoti Bagadthey. 2020. A semi-supervised stacked autoencoder approach for network traffic classification. In *IEEE International Conference on Network Protocols (ICNP)*. 1–6.
- [4] David Berthelot, Nicholas Carlini, Ian Goodfellow, Nicolas Papernot, Avital Oliver, and Colin A. Raffel. 2019. Mixmatch: a holistic approach to semi-supervised learning. In *Neural Information Processing Systems (NIPS)*.
- [5] Qi Cai, Yingwei Pan, Chong-Wah Ngo, Xinmei Tian, Lingyu Duan, and Ting Yao. 2019. Exploring object relation in mean teacher for cross-domain detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 11457–11466.
- [6] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. 2019. Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509* (2019).
- [7] Wenhui Cui, Yanlin Liu, Yuxing Li, Menghao Guo, Yiming Li, Xiuli Li, Tianle Wang, Xiangzhu Zeng, and Mixmatch Ye. 2019. Semi-supervised brain lesion segmentation with an adapted mean teacher model. In *International Conference on Information Processing in Medical Imaging*. 554–565.
- [8] Zihang Dai, Guokun Lai, Yiming Yang, and Quoc Le. 2020. Funnel-Transformer: filtering out sequential redundancy for efficient language processing. In *Neural Information Processing Systems (NIPS)*.
- [9] Zihang Dai, Zhilin Yang, Yiming Yang, Jaime G. Carbonell, Quoc Viet Le, and Ruslan Salakhutdinov. 2019. Transformer-XL: attentive language models beyond a fixed-length context. , 2978–2988 pages.
- [10] Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Lukasz Kaiser. 2019. Universal Transformers. In *International Conference on Learning Representations (ICLR)*.
- [11] Adil Fahad, Abdulmohsen Almalawi, Zahir Tari, Kurayman Alharthi, Fawaz S. Al-Qahtani, and Mohamed Cheriet. 2019. SemTra: A semi-supervised approach to traffic flow labeling with minimal human effort. *Pattern Recognition* 91 (2019), 1–12.
- [12] Angela Fan, Thibaut Lavril, Edouard Grave, Armand Joulin, and Sainbayar Sukhbaatar. 2020. Addressing some limitations of transformers with feedback memory. *arXiv preprint arXiv:2002.09402* (2020).
- [13] Chuanpu Fu, Qi Li, Meng Shen, and Ke Xu. 2021. Real-time robust malicious traffic detection via frequency domain analysis. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 3431–3446.
- [14] Sascha Grollmisch and Estefania Cano. 2021. Improving semi-supervised learning for audio classification with FixMatch. *Electronics* 10, 15 (2021), 1807.
- [15] Qipeng Guo, Xipeng Qiu, Pengfei Liu, Xiangyang Xue, and Zheng Zhang. 2020. Multi-scale self-attention for text classification. In *Conference on Artificial Intelligence (AAAI)*, Vol. 34. 7847–7854.
- [16] Arash Habibi Lashkari, Gerard Draper Gil, Mohammad Mamun, and Ali Ghorbani. 2016. Characterization of encrypted and VPN traffic using time-related features. In *International Conference on Information Systems Security and Privacy*. 407–414.
- [17] Arash Habibi Lashkari, Gerard Draper Gil, Mohammad Saiful Islam Mamun, and Ali A. Ghorbani. 2017. Characterization of Tor traffic using time based features. In *International Conference on Information Systems Security and Privacy*. 253–262.
- [18] Auwal Sani Ilyasu and Huifang Deng. 2019. Semi-supervised encrypted traffic classification with deep convolutional generative adversarial networks. *IEEE Access* 8 (2019), 118–126.
- [19] Chang Liu, Longtao He, Gang Xiong, Zigang Cao, and Zhen Li. 2019. FS-Net: a flow sequence network for encrypted traffic classification. In *IEEE International Conference on Computer Communications (INFOCOM)*. 1171–1179.
- [20] Junming Liu, Yanjie Fu, Jingci Ming, Yong Ren, Leilei Sun, and Hui Xiong. 2017. Effective and real-time in-App activity analysis in encrypted internet traffic streams. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. 335–344.
- [21] Alok Madhukar and Carey Williamson. 2006. A longitudinal study of p2p traffic classification. In *IEEE International Symposium on Modeling, Analysis, and Simulation*. 179–188.
- [22] Yisroel Mirsky, Tomer Doitshman, Yuval Elovici, and Asaf Shabtai. 2018. Kitsune: an ensemble of autoencoders for online network intrusion detection. In *Network and Distributed System Security Symposium (NDSS)*.
- [23] Antonio Montieri, Domenico Ciuonzo, Giuseppe Aceto, and Antonio Pescapé. 2020. Anonymity services Tor, I2P, JonDonym: classifying in the dark (web). *IEEE Transactions on Dependable and Secure Computing* 17, 3 (2020), 662–675.
- [24] Eva Papadogiannaki and Sotiris Ioannidis. 2021. A survey on encrypted network traffic analysis applications, techniques, and countermeasures. *Comput. Surveys* 54, 6 (2021).
- [25] Niki Parmar, Ashish Vaswani, Jakob Uszkoreit, Lukasz Kaiser, Noam Shazeer, Alexander Ku, and Dustin Tran. 2018. Image Transformer. In *International Conference on Machine Learning (ICML)*. 4055–4064.
- [26] Shahbaz Rezaei and Xin Liu. 2019. Deep learning for encrypted traffic classification: an overview. *IEEE Communications Magazine* 57, 5 (2019), 76–81.
- [27] Shahbaz Rezaei and Xin Liu. 2019. How to achieve high classification accuracy with just a few labels: A semi-supervised approach using sampled packets. In *Industrial Conference on Data Mining*. 1–15.
- [28] Meng Shen, Yiting Liu, Liehuang Zhu, Ke Xu, Xiaojiang Du, and Nadra Guizani. 2020. Optimizing feature selection for efficient encrypted traffic classification: a systematic approach. *IEEE Network* 34, 4 (2020), 20–27.
- [29] Kihyuk Sohn, David Berthelot, Nicholas Carlini, Zizhao Zhang, Han Zhang, Colin Raffel, Ekin Dogus Cubuk, Alexey Kurakin, and Chun-Liang Li. 2020. FixMatch: simplifying semi-supervised learning with consistency and confidence. In *Neural Information Processing Systems (NIPS)*.
- [30] Antti Tarvainen and Harri Valpola. 2017. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. In *International Conference on Learning Representations (ICLR)*.
- [31] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Neural Information Processing Systems (NIPS)*. 5998–6008.
- [32] Vikas Verma, Kenji Kawaguchi, Alex Lamb, Juho Kannala, Arno Solin, Yoshua Bengio, and David Lopez-Paz. 2022. Interpolation consistency training for semi-supervised learning. *Neural Networks* 145 (2022), 90–106.
- [33] Benyou Wang, Donghao Zhao, Christina Lioma, Qiuchi Li, Peng Zhang, and Jakob Grue Simonsen. 2019. Encoding word order in complex embeddings. In *International Conference on Learning Representations (ICLR)*.
- [34] Jing Wang, Jian Tang, Zhiyuan Xu, Yanzhi Wang, Guoliang Xue, Xing Zhang, and Dejun Yang. 2017. Spatiotemporal modeling and prediction in cellular networks: a big data enabled deep learning approach. In *IEEE International Conference on Computer Communications (INFOCOM)*. 1–9.
- [35] Lingfeng Wang, Shisen Wang, Jin Qi, and Kenji Suzuki. 2021. A multi-task mean teacher for semi-supervised facial affective behavior analysis. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 3603–3608.
- [36] Wei Wang, Ming Zhu, Xuewen Zeng, Xiaozhou Ye, and Yiqiang Sheng. 2017. Malware traffic classification using convolutional neural network for representation learning. In *International Conference on Information Networking*. 712–717.
- [37] Zhanghao Wu, Zhijian Liu, Ji Lin, Yujun Lin, and Song Han. 2019. Lite transformer with long-short range attention. In *International Conference on Learning Representations (ICLR)*.
- [38] Gaogang Xie, Kun Xie, Jun Huang, Xin Wang, Yuxiang Chen, and Jigang Wen. 2017. Fast low-rank matrix approximation with locality sensitive hashing for quick anomaly detection. In *IEEE International Conference on Computer Communications (INFOCOM)*. 1–9.
- [39] Yu Yan, Lin Qi, Jie Wang, Yun Lin, and Lei Chen. 2020. A network intrusion detection method based on stacked autoencoder and LSTM. In *IEEE International Conference on Communications (ICC)*. 1–6.
- [40] Ying Yang, Cuicui Kang, Gaopeng Gou, Zhen Li, and Gang Xiong. 2018. TLS/SSL encrypted traffic classification with autoencoder and convolutional neural network. In *IEEE International Conference on High Performance Computing and Communications*. 362–369.
- [41] Haipeng Yao, Pengcheng Gao, Peiying Zhang, Jingjing Wang, Chunxiao Jiang, and Lijun Lu. 2019. Hybrid intrusion detection system for edge-based IIoT relying on machine-learning-aided detection. *IEEE Network* 33, 5 (2019), 75–81.
- [42] Haipeng Yao, Chong Liu, Peiying Zhang, Sheng Wu, Chunxiao Jiang, and Shui Yu. 2019. Identification of encrypted traffic through attention mechanism based long short term memory. *IEEE Transactions on Big Data* (2019), 1–1.
- [43] Yi Zeng, Huaxi Gu, Wenting Wei, and Yantao Guo. 2019. Deep-Full-Range: a deep learning based network encrypted traffic classification and intrusion detection framework. *IEEE Access* (2019), 45182–45190.
- [44] Hongyi Zhang, Moustapha Cissé, Yann N. Dauphin, and David Lopez-Paz. 2018. mixup: beyond empirical risk minimization. In *International Conference on Learning Representations (ICLR)*.
- [45] Jielun Zhang, Fuhao Li, Feng Ye, and Hongyu Wu. 2020. Autonomous unknown-application filtering and labeling for DL-based traffic classifier update. In *IEEE International Conference on Computer Communications (INFOCOM)*. 397–405.
- [46] Ruijie Zhao, Yiteng Huang, Xianwen Deng, Zhi Xue, Jiabin Li, Zijiang Huang, and Yijun Wang. 2021. Flow Transformer: a novel anonymity network traffic classifier with attention mechanism. In *International Conference on Mobility, Sensing and Networking (MSN)*. 1–8.
- [47] Wenbo Zheng, Chao Gou, Lan Yan, and Shaocong Mo. 2020. Learning to classify: a flow-based relation network for encrypted traffic classification. In *The Web Conference (WWW)*. 13–22.
- [48] Dengyong Zhou, Olivier Bousquet, Thomas Lal, Jason Weston, and Bernhard Schölkopf. 2004. Learning with local and global consistency. In *Neural Information Processing Systems (NIPS)*, Vol. 16.